WATT THE HACK

PowerDrone

NOMADIC DRONES CHALLENGE: KEEPING THE POWER FLOWING

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Problem Definition



Develop a plan for using mobile, self-charging sensors that can fly between power lines to investigate grid damage. Decide what to measure, where to deploy limited sensors and how new findings guide their next move—and when there is enough evidence to recommend an inspection or repair.

Available Data & Information

What data feeds our prioritization

Background and live data rank the areas. Drones check the top ones and their findings update the ranking.

BASELINE · STATIC

Background data

Environment

Terrain and exposure, forest near lines, flood risk, climate and season

Network

Topology and redundancy, criticality, asset vulnerability

Consequence

Population density, critical facilities

Past experience

Past problems and actions taken, operational anomalies

EVIDENCE · AREA LEVEL

Live event data

Every report carries

ID, type, description, location or asset, time relative to storm, confidence, source

Measured data

Wind and gusts, storm warnings, lightning, temperature, grid system logs

Note: not every type of live data is available in every event.

VERIFICATION · ASSET LEVEL

Drone inspection data

Core sensors

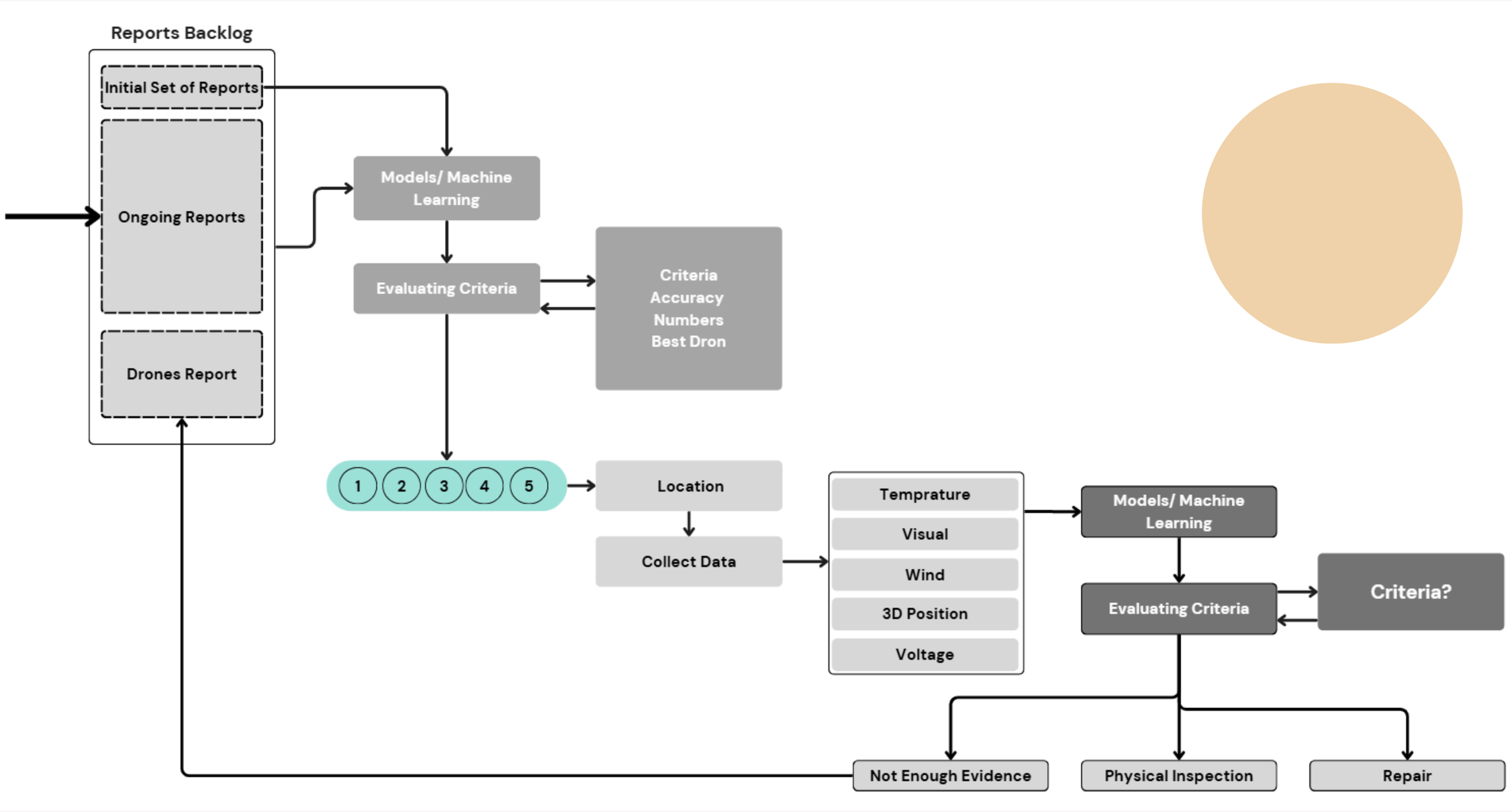
Current probe, thermal camera, RGB camera and LiDAR, IMU for sag and swing

Add-ons when needed

Ultrasound and UV corona camera, gas sensor, ice load sensor

Drones measure the status of all six storm problems: trees, damaged wires, low lines, wind swing, ice and fire.

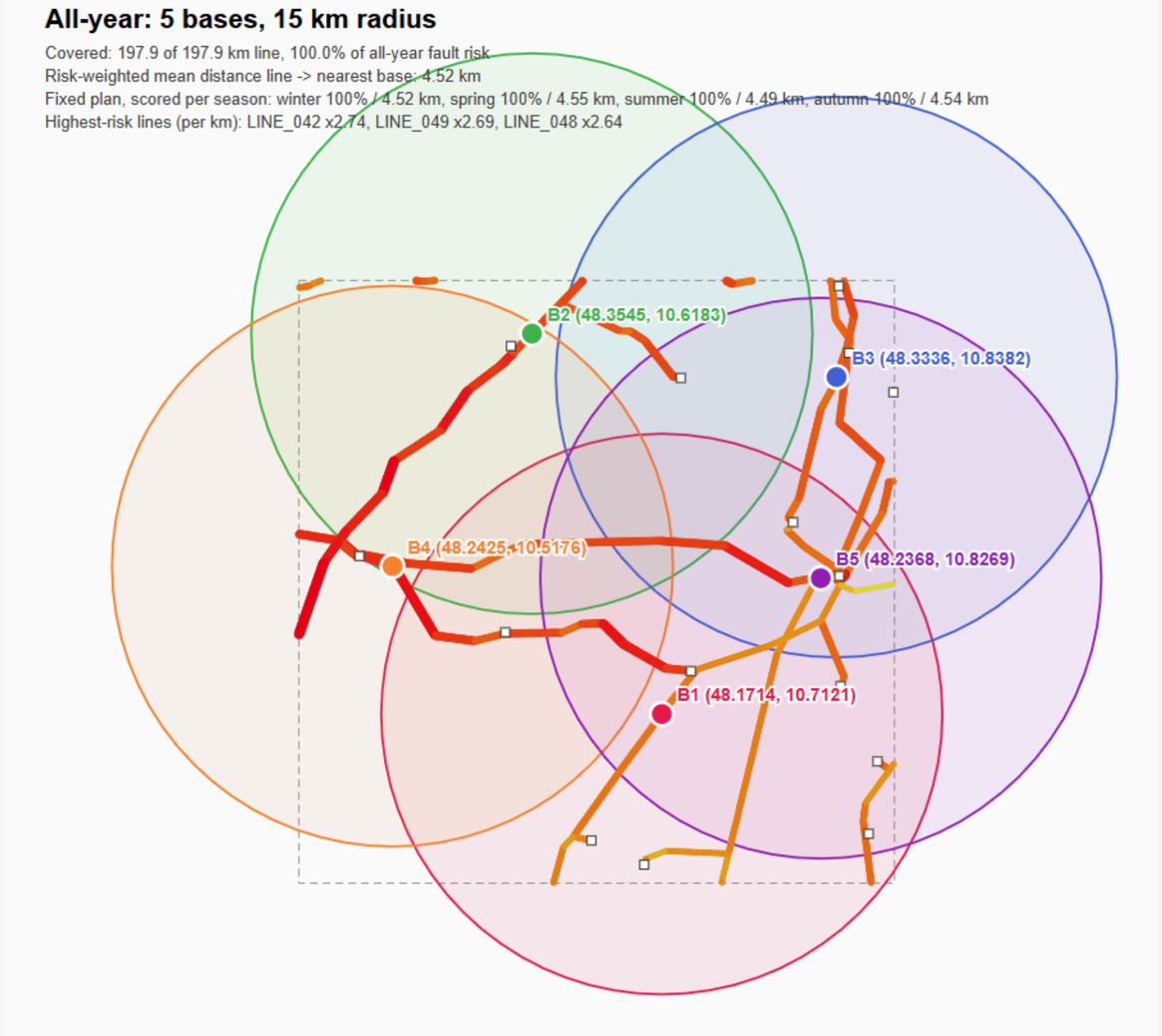
Algorithm


Reports Backlog
Initial Set of Reports
Ongoing Reports
Drones Report
Models/ Machine Learning
Evaluating Criteria
Criteria
Accuracy
Numbers
Best Dron
1
2
3
4
5
Location
Collect Data
Temprature
Visual
Wind
3D Position
Voltage
Models/ Machine Learning
Evaluating Criteria
Criteria?
Not Enough Evidence
Physical Inspection
Repair

Initial Placement Point

Initial Placement Point

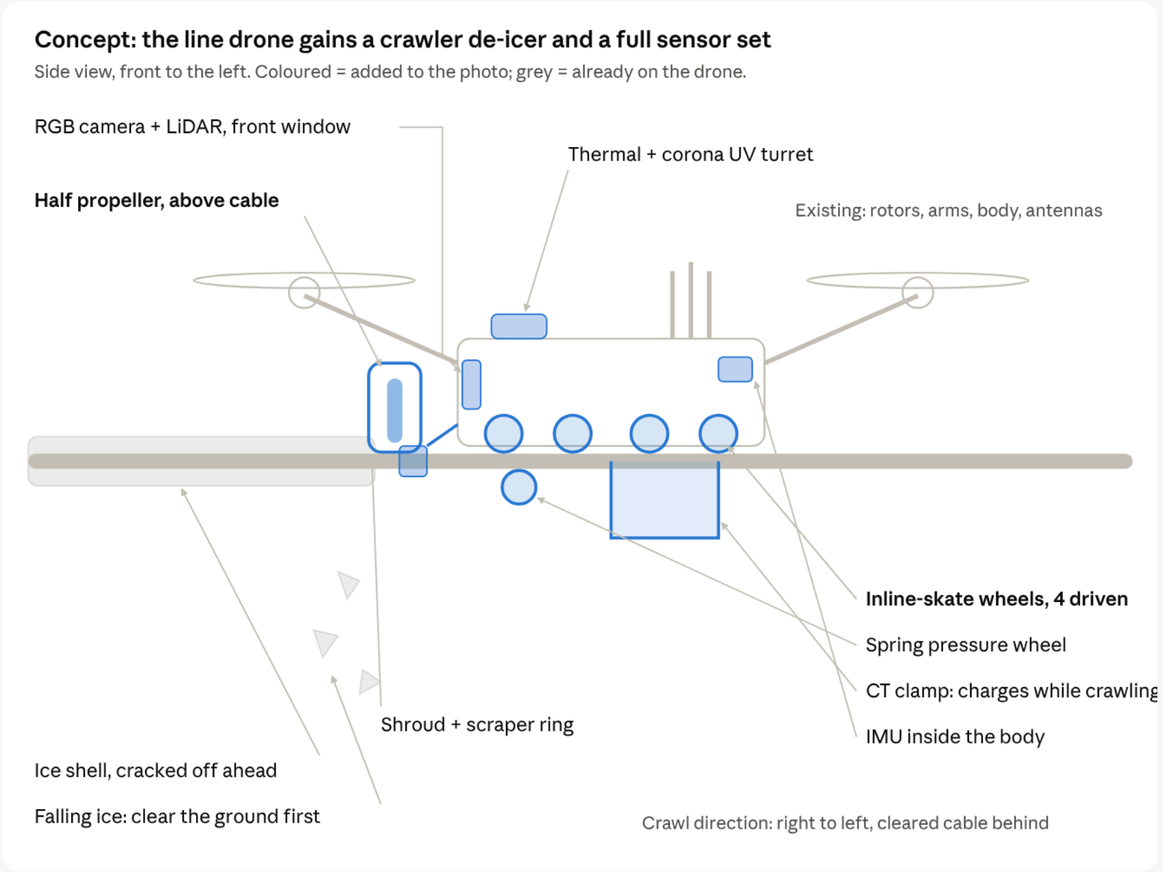
- Coverage Calculation: Measures the power-line length within radius R of each candidate base and weights it by fault probability.
- Base Selection: Greedily selects K bases maximizing additional weighted coverage, then improves placement through 1-swap optimization.
- Distance Optimization: Applies p-median optimization to move bases closer to power lines without reducing coverage.
- Seasonal Risk Adjustment: Recalculates fault-risk weights based on wind, ice, vegetation, lightning, and heat, while excluding de-energized lines from base placement.



Hardware Innovation

Active De-Icing





The concept enhances the existing Nomadic Drone with a line-crawling mechanism powered directly by the power line, enabling continuous movement and inspection along electrical cables. A front-mounted rotating propeller continuously adjusts its direction to mechanically break and remove accumulated ice as the drone moves along the line. This combines autonomous inspection, self-powered crawling, and active de-icing to improve power grid resilience during extreme weather conditions.

Thank You!